

Prerequisites: Log in to HUAWEI CLOUD.

[Show less](#)

What is the difference between a HUAWEI CLOUD experiment account provided by the system and a real HUAWEI CLOUD account?

Access Exercise Interface, open Google Chrome and access the HUAWEI CLOUD login page. Select IAM User Login. In the login dialog box, enter the HUAWEI CLOUD lab account and password allocated by the system to log in to HUAWEI CLOUD, as shown in the following figure.

The image displays two side-by-side login interfaces. The left interface is titled 'HUAWEI ID login' and contains a form with two input fields: 'Phone/Email/Login ID/HUAWEI CLOUD account name' and 'Password'. Below these fields is a red 'LOG IN' button. Underneath the button are links for 'Register' and 'Forgot password'. At the bottom, there is a section titled 'Use Another Account' with a list of options: 'IAM User', 'Federated User', 'Huawei Website Account', 'Huawei Enterprise Partner', and 'HUAWEI CLOUD Account'. The 'IAM User' option is highlighted with a red box. The right interface is titled 'IAM User Login' and contains a form with three input fields: 'Tenant name or HUAWEI CLOUD account name', 'IAM user name or email address', and 'IAM user password'. Below these fields is a red 'Log In' button. At the bottom, there are links for 'Forgot Password' and a 'Remember me' checkbox. A red arrow points from the 'IAM User' link in the left interface to the 'IAM User Login' form on the right.

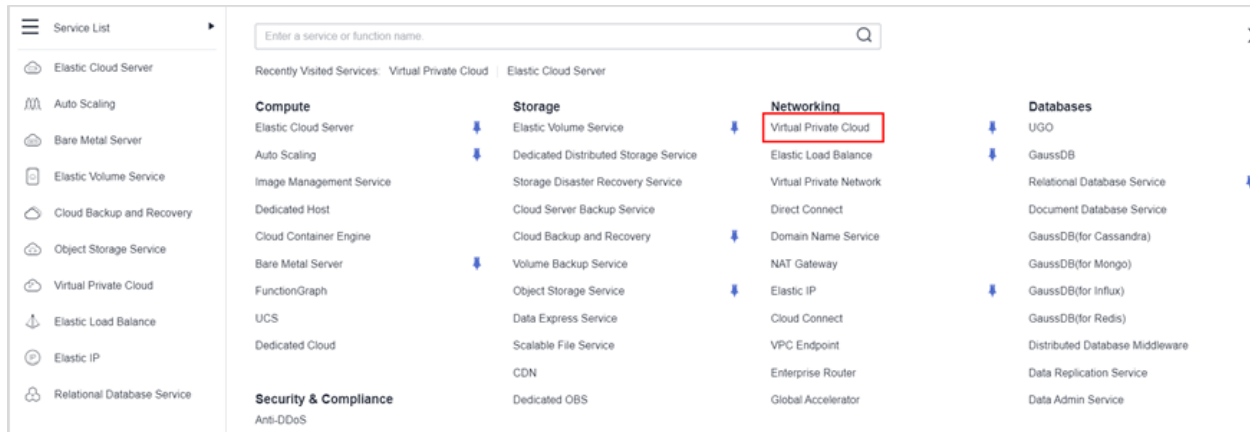
Note: For details about the account information, see the upper part of the lab manual. Do not use your HUAWEI CLOUD account to log in.

1 Monitoring ECS Performance

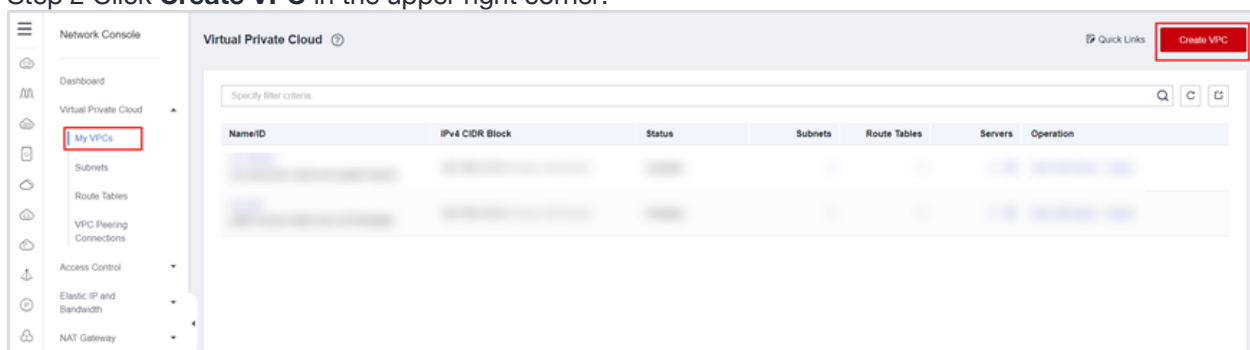
[Show less](#)

1.1 Purchasing an ECS

Step 1 Log in to the Huawei Cloud console, click **Service List** in the upper left corner, and choose **Networking > Virtual Private Cloud**.



Step 2 Click **Create VPC** in the upper right corner.



Step 3 On the **Create VPC** page, configure the parameters as follows:

-- **Region: AP-Singapore**

-- **Name: vpc-test**

-- **Name of Default Subnet: subnet-test**

Retain the default values for other parameters.

Click **Create Now**.

×

Create Security Group

★ Name

sg-test

★ Template

General-purpose web server

Description

The security group is for general-purpose web servers and includes default rules that allow all inbound ICMP traffic and inbound traffic on ports 22, 80, 443, and 3389. The security group is used for remote login, ping, and hosting a website on ECSs.

0/255

Show Default Rule

OK

Cancel

Step 6 Back to **Service List**, and choose **Compute > Elastic Cloud Server**.

HUAWEI CLOUD

Console

Hong Kong

Service List

Elastic Cloud Server

Auto Scaling

Bare Metal Server

Elastic Volume Service

Cloud Backup and Recovery

Object Storage Service

Virtual Private Cloud

Elastic Load Balance

Compute

Elastic Cloud Server

Auto Scaling

Image Management Service

Dedicated Host

Cloud Container Engine

Bare Metal Server

FunctionGraph

UCS

Storage

Elastic Volume Service

Dedicated Distributed Storage Service

Storage Disaster Recovery Service

Cloud Server Backup Service

Cloud Backup and Recovery

Volume Backup Service

Object Storage Service

Data Express Service

Networking

Virtual Private Cloud

Elastic Load Balance

Virtual Private Network

Direct Connect

Domain Name Service

NAT Gateway

Elastic IP

Cloud Connect

Step 7 Click **Buy ECS** in the upper right corner. In the displayed page, configure the parameters as follows:

- **Billing Mode: Pay-per-use**
- **Region: AP-Singapore**
- **AZ: Random**

- CPU Architecture: x86
- Specifications: c6.large.2 2 vCPUs | 4 GiB
- Image: Public image, and CentOS 7.6 64bit(40 GB)
- System Disk: High I/O, and 40 GB

Retain the default values for other parameters.

The screenshot shows the 'Elastic Cloud Server' configuration interface. The progress bar at the top indicates four steps: 1. Configure Basic Settings (active), 2. Configure Network, 3. Configure Advanced Settings, and 4. Confirm.

Billing Mode: Yearly/Monthly, **Pay-per-use** (selected), Spot price.

Region: CN-Hong Kong (selected). Below it, a note says: 'For low network latency and quick resource access, select the region nearest to your target users. Learn how to select a region.'

AZ: Random (selected), AZ1, AZ2, AZ3.

CPU Architecture: x86 (selected), Kunpeng.

Specifications: Latest generation, vCPUs: All, Memory: All, Flavor Name: [search bar].

General computing-plus (selected tab). Other tabs: General computing, Memory-optimized, Large-memory, High-performance computing, Disk-intensive, Ultra-high I/O, GPU-accelerated, AI-accelerated.

Flavor Name	vCPUs Memory(GiB)	CPU	Assured / Maximum Bandwidth	Packets Per Second (PPS)
☒ c6.large.2	2 vCPUs 4 GiB	Intel Cascade Lake 3.0GHz	1.2 / 4 Gbit/s	400,000
☐ c6.large.4	2 vCPUs 8 GiB	Intel Cascade Lake 3.0GHz	1.2 / 4 Gbit/s	400,000
☐ c6.xlarge.2	4 vCPUs 8 GiB	Intel Cascade Lake 3.0GHz	2.4 / 8 Gbit/s	800,000
☐ c6.xlarge.4	4 vCPUs 16 GiB	Intel Cascade Lake 3.0GHz	2.4 / 8 Gbit/s	800,000
☐ c6.2xlarge.2	8 vCPUs 16 GiB	Intel Cascade Lake 3.0GHz	4.5 / 15 Gbit/s	1,500,000
☐ c6.2xlarge.4	8 vCPUs 32 GiB	Intel Cascade Lake 3.0GHz	4.5 / 15 Gbit/s	1,500,000

Image: Public image (selected), Private image, Shared image, Marketplace image.

OS: CentOS (selected), CentOS 7.6 64bit(40GB) (selected).

Host Security: ☒ Enable, Basic (free) (selected).

System Disk: High I/O (selected), 40 (selected) GiB. IOPS limit: 2,120, IOPS burst limit: 5,000.

Quantity: 1. **ECS Price:** \$0.101 USD/hour. This price is an estimate and may differ from the final price. [Pricing details](#).

Next: Configure Network

Step 8 Click **Next: Configure Network** and configure the parameters as follows:

- Network: vpc-test, and subnet-test
- Security Group: sg-test
- EIP: Auto Assign
- EIP Type: Dynamic BGP

Elastic Cloud Server

1 Configure Basic Settings 2 **Configure network** 3 Configure Advanced Settings 4 Confirm

Network: **sg-test (192.168.0.0/16)** | **subnet-test (192.168.0.0/24)** | **Automatically assign IP address** | Available private IP addresses: 250

Extension NIC: Add NIC. NICs you can still add: 1

Security Group: **sg-test (4b5b0a7b-87be-40d1-b2c2-0b037fa7b32b)** | Create Security Group

Similar to a firewall, a security group logically controls network access.
Ensure that the selected security group allows access to port 22 (SSH-based Linux login), 3389 (RDP Windows login), and ICMP (ping operations). Configure Security Group Rules

Security Group Rules

Security Group Name	Priority	Action	Protocol & Port	Type	Source	Description
sg-test	1	Permit	ICMP: All	IPv4	0.0.0.0/0	允許ping程序測試彈性云服務器的连通性
	1	Permit	TCP: 22	IPv4	0.0.0.0/0	允許SSH這種遠端Linux操作系統登錄
	1	Permit	TCP: 443	IPv4	0.0.0.0/0	允許使用HTTPS协议访问网站
	1	Permit	TCP: 3389	IPv4	0.0.0.0/0	允許遠程登錄Windows操作系統登錄
	1	Permit	All	IPv4	sg-test	允許安全組內的彈性云服務器間的通信

IP: ☒ Auto assign ☐ Use existing ☐ Not required

IP Type: **Dynamic BGP** | Premium BGP

Greater than or equal to 99.95% service availability rate

Billed By: **Bandwidth** | Traffic | Shared bandwidth

Bandwidth Size: 1 | 2 | **5** | 10 | 100 | 200 | Custom | The bandwidth can be from 1 to 500 Mbit/s.

Free Anti-DDoS protection

Quantity: 1 | ECS Price: **\$0.151 USD/hour**

Previous **Next: Configure Advanced Settings**

Step 9 Click **Next: Configure Advanced Settings**.

-- **ECS Name: ecs-test**

-- **Login Mode: Password**

-- **Password:** Customize a password, for example, Huawei@1(You are advised to set a complex password.)

-- **Cloud Backup and Recovery: Not required**

Retain the default values for other parameters and click Next: Confirm.

1 Configure Basic Settings — 2 Configure Network — 3 Configure Advanced Settings — 4 Confirm

ECS Name: ☐ Allow duplicate name

If multiple ECSs are created at the same time, the system automatically adds a hyphen followed by a four-digit incremental number to the end of each ECS name. For example, if you enter ecs and there is no existing ECS in the system, the first ECS's name will be ecs-0001. If an ECS with the name ecs-0010 already exists, the name of the first new ECS will be ecs-0011.

Login Mode: ☐ Key pair ☒ Password ☐ Set password later

Username: root

Password: Keep the password secure. If you forget the password, you can log in to the ECS console and change it.

Confirm Password:

Cloud Backup and Recovery: ☐ Create new ☐ Use existing ☒ Not required ?

CBR backups can help you restore data in case anything happens to your ECS. To ensure data security, you are advised to use CBR.

Cloud Eye: ☒ Enable Detailed Monitoring Free ?

Enable 1-minute fine-grained monitoring of ECS metrics, such as CPU, memory, network, disk, and process.

ECS Group (Optional): ?

?

[Create ECS Group](#)

Advanced Options: ☐ Configure now

Quantity:

ECS Price: **\$0.151 USD/hour**
This price is an estimate and may differ from the final price. [Pricing details](#)

Step 10 Select the agreement and click **Submit**.

1 Configure Basic Settings — 2 Configure Network — 3 Configure Advanced Settings — 4 Confirm

Configuration

Basic		Network		Advanced	
Billing Mode	Pay per-use	VPC	vpc-test (192.168.0.0/16)	ECS Name	ecs-test
Specifications	General computing-plus c6.large.2 2 vCPUs 4 GiB	EIP	Dynamic BGP Bill by Bandwidth Bandwidth: 5 Mbit/s	Login Mode	Password
System Disk	High I/O, 40 GiB	Security Group	sg-test	ECS Group	--
Region	Hong Kong	Primary NIC	subnet-test (192.168.0.0/24)	Cloud Eye	Monitoring details
Image	CentOS 7.6 64bit				
AZ	Host Security				
AZ2	Basic (Free)				

Launch Template: ?

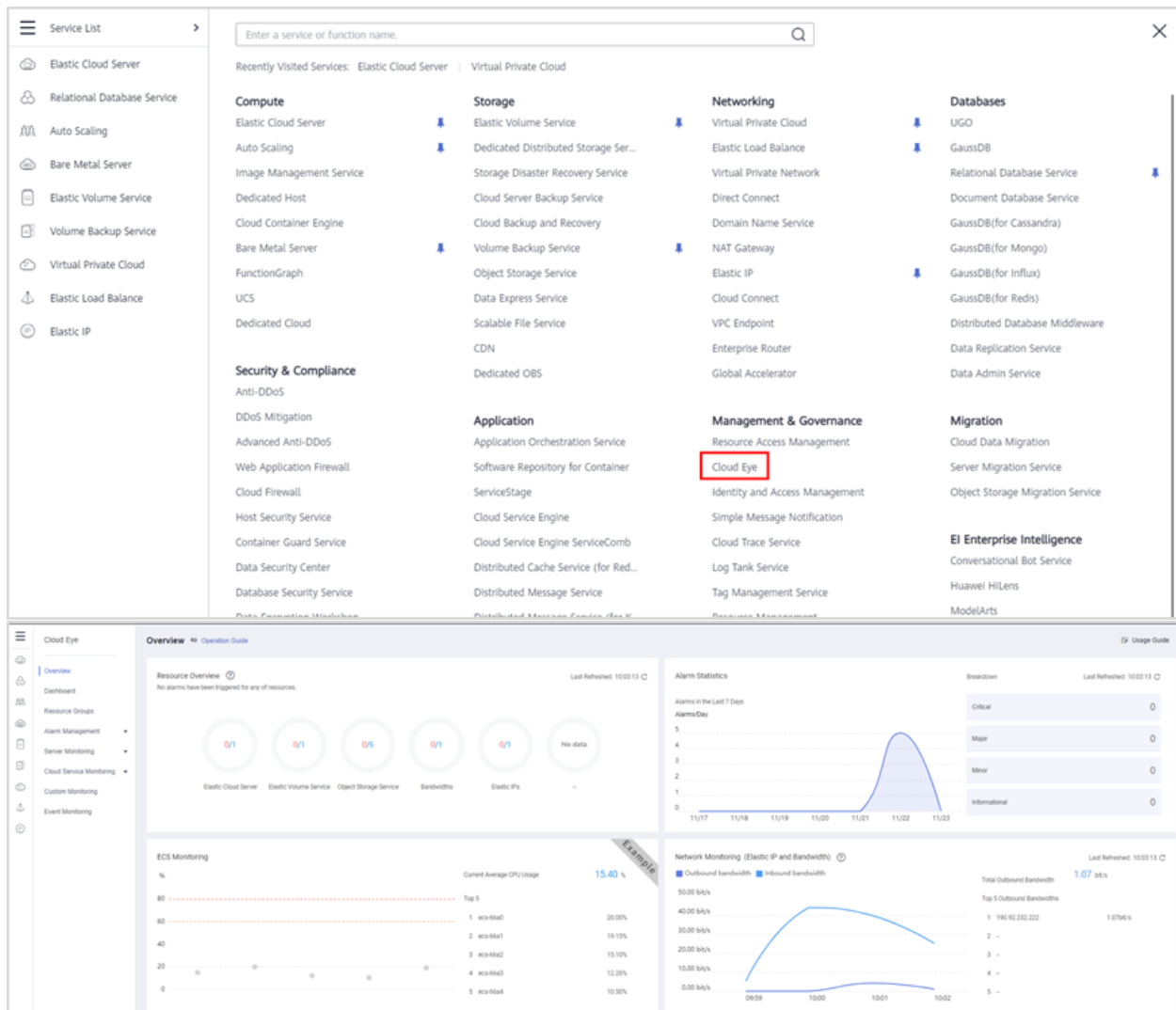
Quantity: You can create a maximum of 20 ECSs. [Learn how to increase quota.](#)

Agreement: ☒ I have read and agree to the Service Level Agreement and Image Disclaimer.

ECS Price: **\$0.151 USD/hour**
This price is an estimate and may differ from the final price. [Pricing details](#)

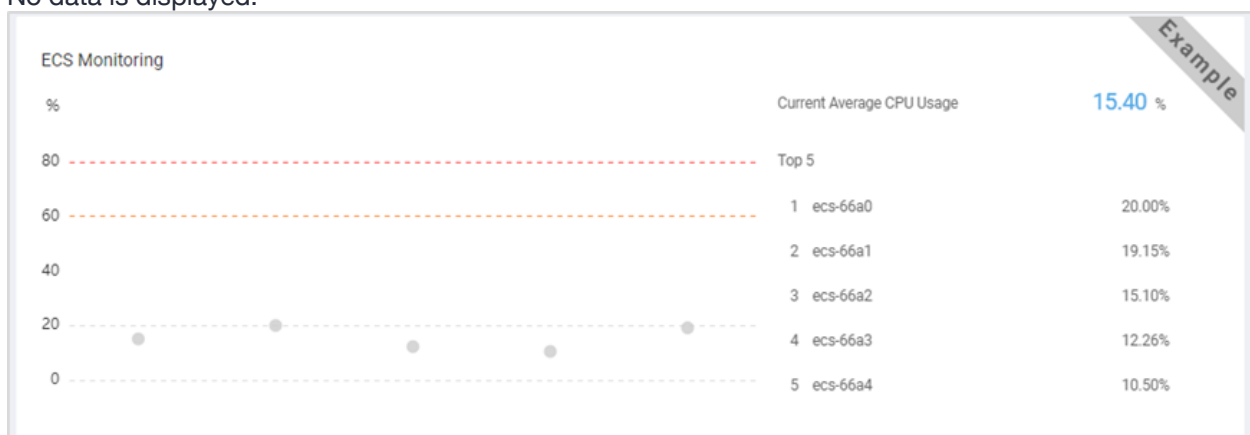
1.2 Using Cloud Eye

Step 1 Log in to the Huawei Cloud console, click **Service List**, and choose **Management & Governance > Cloud Eye**.

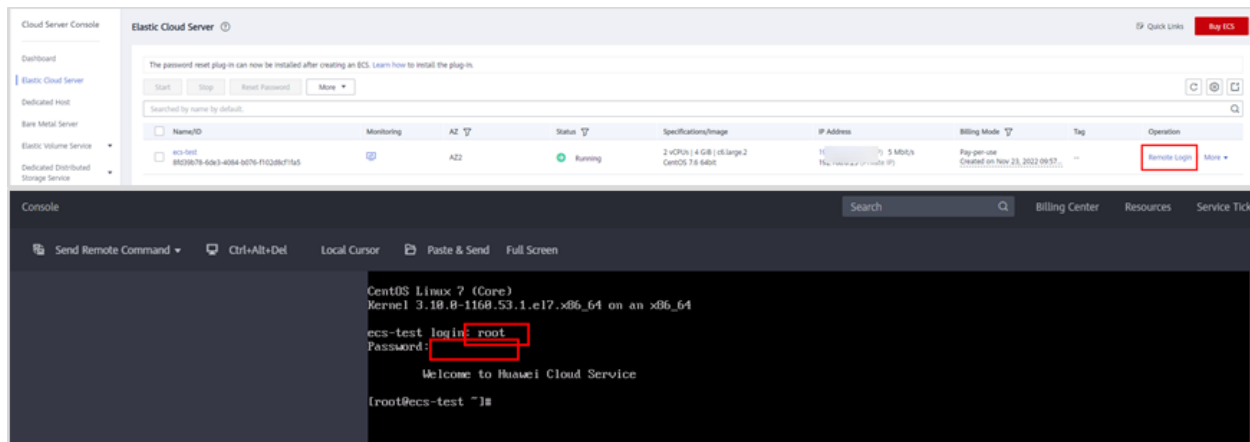


Step 2 On the **Overview** page, check **ECS Monitoring**.

No data is displayed.



Step 3 On the **Elastic Cloud Server** page, locate **ecs-test**, and click **Remote Login** to log in to **ecs-test** and install the Cloud Eye Agent. Wait for 3 to 5 minutes until the Agent collects monitoring data.



```
cd /usr/local && curl -k -O https://obs.cn-north-4.myhuaweicloud.com/uniagent-cn-north-4/script/agent_install.sh && bash agent_install.sh
```

```
[root@ecs-test ~]# cd /usr/local && curl -k -O https://obs.cn-north-4.myhuaweicloud.com/uniagent-cn-north-4/script/agent_install.sh && bash agent_install.sh
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 3833 100 3833 0 0 0 0:02:12 0:02:10 0:00:02 865
```

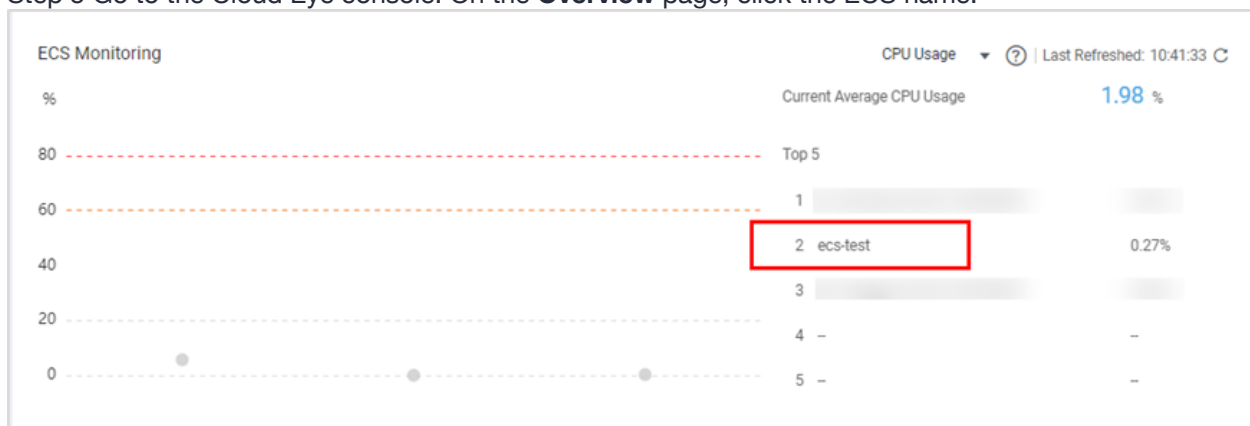
Step 4 Run the following command on the ECS CLI to use the shell script:

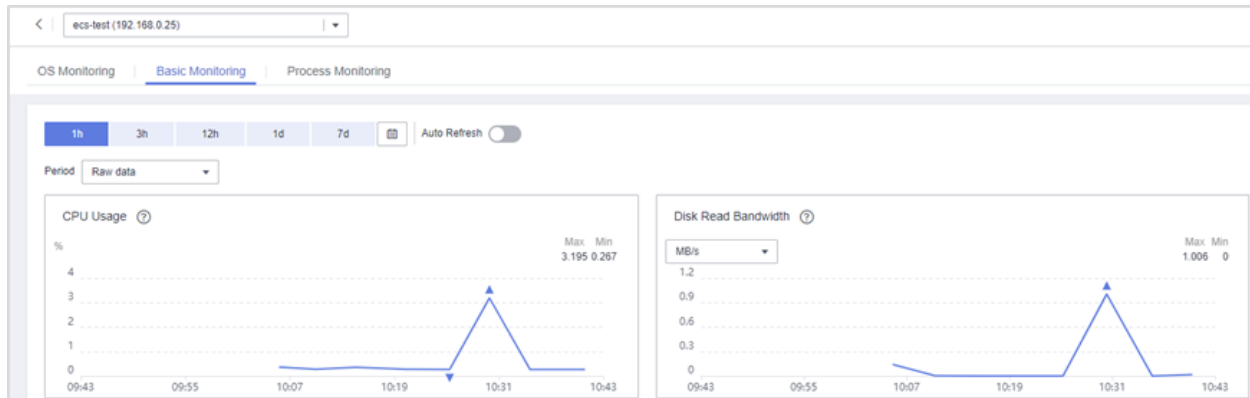
```
vim test.sh
```

```
#!/bin/bash
a=1
while true
do
a=$((a+1))
echo $a
done
sh test.sh
```

```
[root@ecs-test ~]# vim test.sh
[root@ecs-test ~]# sh test.sh
```

Step 5 Go to the Cloud Eye console. On the **Overview** page, click the ECS name.

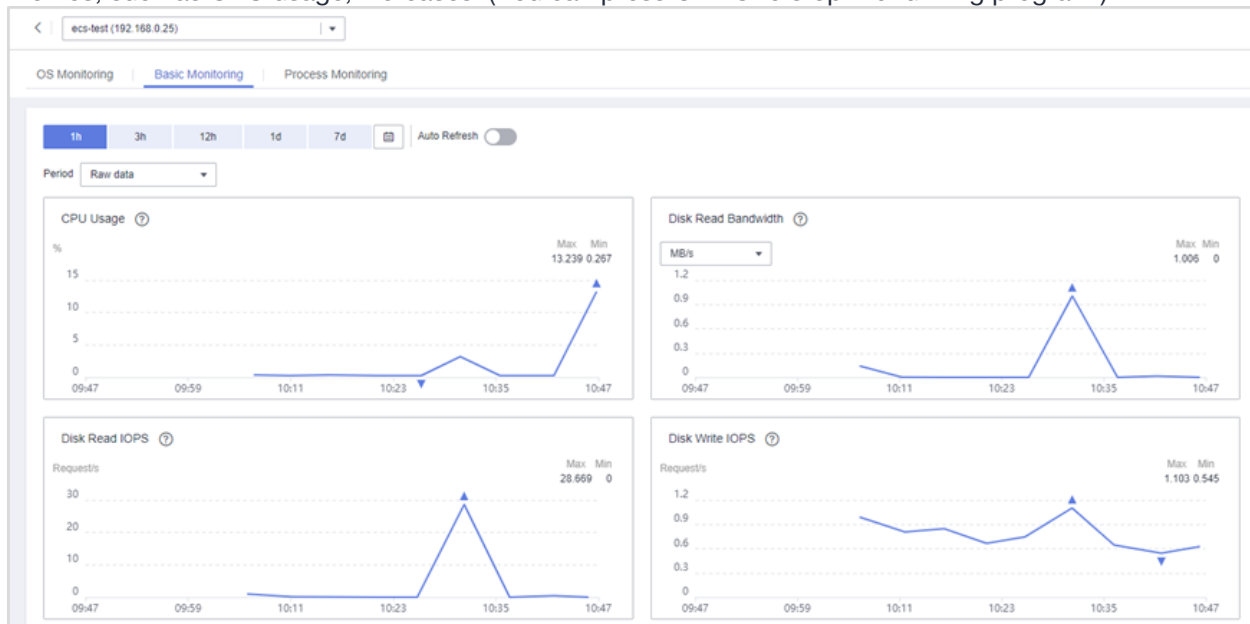




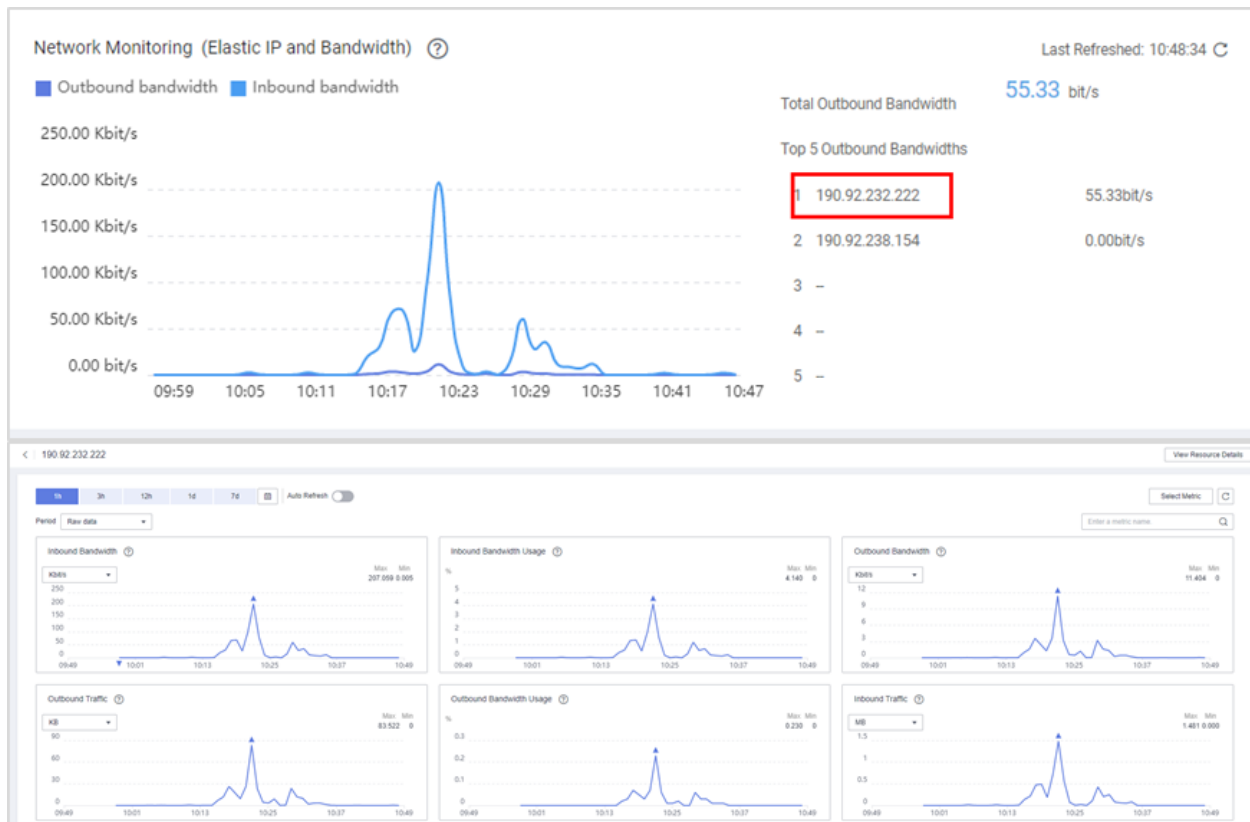
Step 6 Run the following command on the ECS:

```
sh test.sh
```

Step 7 Go back to the **Overview** page, click the ECS name. Check whether monitoring data of the ECS metrics, such as CPU usage, increases. (You can press Ctrl+C to stop the running program.)



Step 8 Go back to the **Overview** page. In the **Network Monitoring (Elastic IP and Bandwidth)** area, click an Elastic IP (EIP) address. On the displayed page, view fluctuations in network traffic.



Step 9 Go back to the **Overview** page. In the **Storage Monitoring (EVS)** area, click the name of the Elastic Volume Service (EVS) disk. On the displayed page, view the fluctuation of EVS disk read and write bandwidth/IOPS.



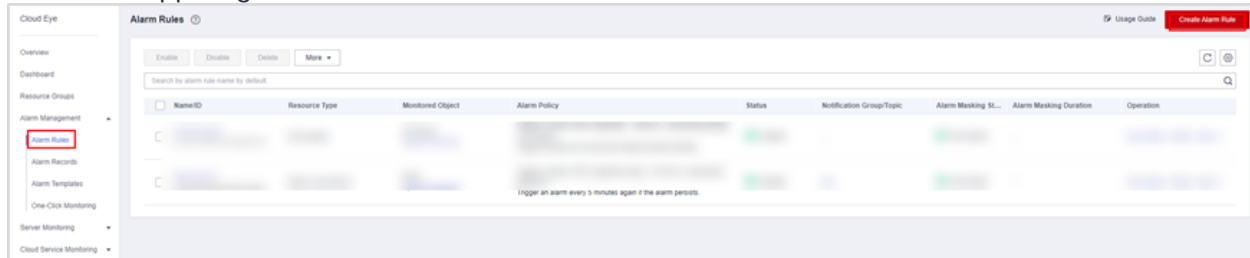


2 Alarms

[Show less](#)

2.1 Creating an Alarm Rule

Step 1 On the Cloud Eye console, choose **Alarm Management > Alarm Rules**, and click **Create Alarm Rule** in the upper right corner.



-- **Name:** alarm-test

-- **Resource Type:** Elastic Cloud Server

-- **Dimension:** ECSs

-- **Monitoring Scope:** Specific resources

Select the created ECS.

-- **Method:** Configure manually

-- **Alarm Policy:** An alarm is generated every 5 minutes if the maximum CPU usage is 5% or more within the monitoring period (5 minutes) for one consecutive period.

-- **Notification Object:** test

-- **Notification Window:** Retain the default setting.

-- **Trigger Condition:** Retain the default setting.

* Name:

Description:

* Resource Type:

* Dimension:

* Monitoring Scope:

Select All

Name	ID
☐ as-config-service01-B7YB4YBP	d831570c-4235-45ed-83f6-3e4af59e5c87
☐ as-config-service01-AHLOS111	7148cf88-25ee-4ac5-958a-59ad07408653
☐ as-config-service01-KV38HN9P	4a582cd9-1861-42d3-bbd6-9acd23dbae9b
☐ ecs-test	8839b78-6de3-4084-b076-f102d8cf1fa5

 >> <<

Deselect All

Name	ID
No data available.	

* Method:

* Alarm Policy

Metric Name	Alarm Policy	Alarm Seve...	Operation
CPU Usage	Max 5 minutes 1 consecuti... >= 5 % Every 5 minutes	Major	

Add Alarm Policy You can add 40 more

Alarm Notification: ☒

* Notification Object:

Create an SMN topic and click refresh to make it available for selection.

* Notification Window: Daily 00:00 - 23:59

* Trigger Condition: ☒ Generated alarm ☒ Cleared alarm

Step 2 Click **Create**. On the **Alarm Rules** page, view the created alarm rule.

Cloud Eye	Alarm Rules	Usage Guide	Create Alarm Rule
Overview	Enable Disable Delete More		
Dashboard	Search by alarm rule name by default.		
Resource Groups			
Alarm Management			
Alarm Rules			
Alarm Records			
Alarm Templates			
One-Click Monitoring			
Server Monitoring			
Cloud Service Monitoring			
Custom Monitoring			
Event Monitoring			

NameID	Resource Type	Monitored Object	Alarm Policy	Status	Notification Gro...	Alarm Ma...	Alarm Masking D...	Operation
alarm-test	Elastic Cloud Server	ECSs	Trigger an alarm if CPU Usage Max >= 5% for 1 consecutive periods of 5 minutes. Trigger an alarm every 5 minutes again if the alarm persists.	Enabled	test	Not masko		View Details Modify More

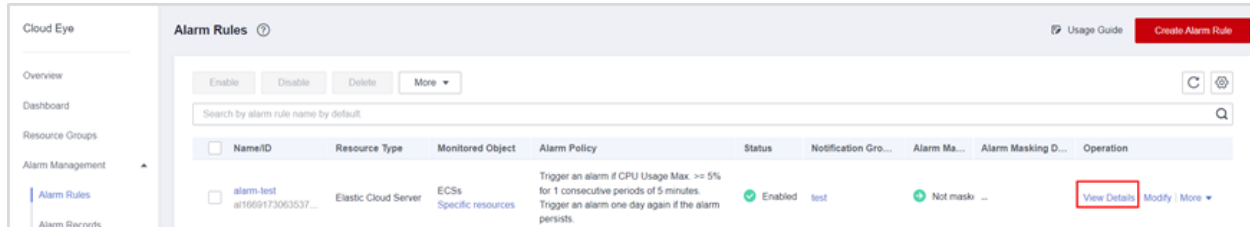
2.2 Triggering an Alarm

Step 1 Run the following command on the CLI:

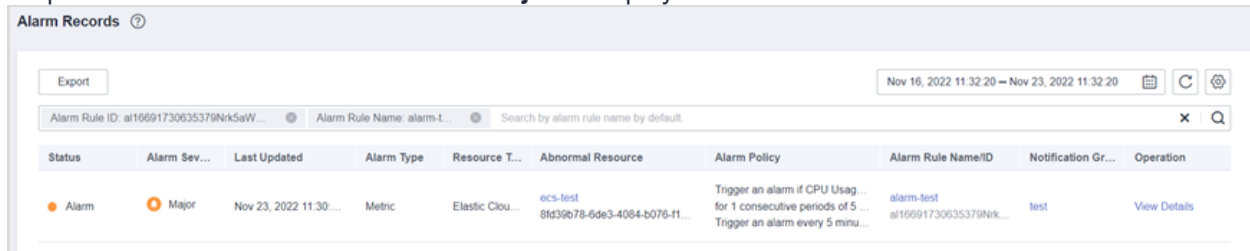
```
sh test.sh
```

```
[root@ecs-test ~]# vim test.sh
[root@ecs-test ~]# sh test.sh
```

Step 2 Wait for 5 minutes, return to the Cloud Eye console, and choose **Alarm Management > Alarm Records**.



Step 3 An alarm record whose status is **Major** is displayed.

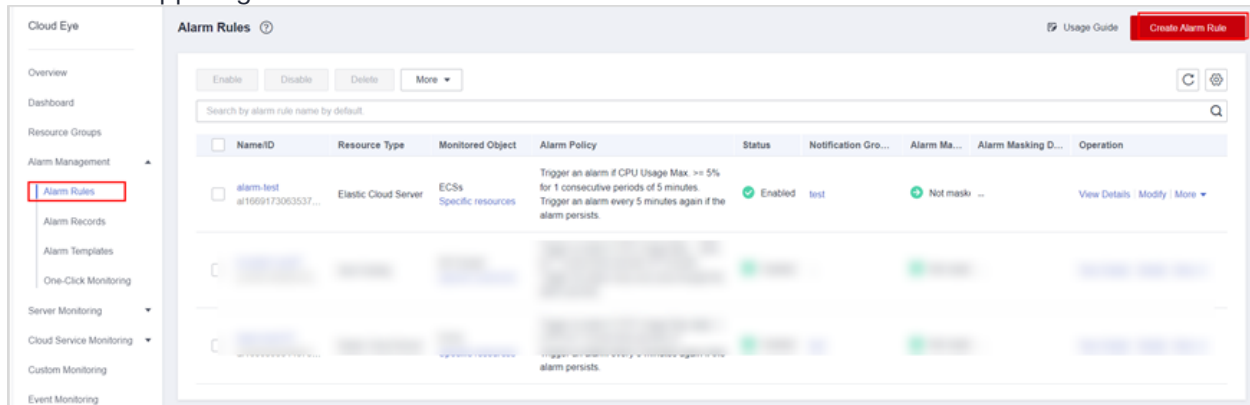


Step 4 Press **Ctrl+Z** to stop the running program.

3 Using Cloud Eye Together with SMN
[Show less](#)

3.1 Creating an Alarm Rule

Step 1 On the Cloud Eye console, choose **Alarm Management > Alarm Rules**, and click **Create Alarm Rule** in the upper right corner.



-- Name: alarm-test2

-- Resource Type: Elastic Cloud Server

-- Dimension: ECSs

-- Monitoring Scope: Specific resources

Select the **created ECS**.

-- **Method: Configure manually**

-- **Alarm Policy:** An alarm is generated every 5 minutes if the maximum memory usage is 5% or more within the monitoring period (5 minutes) for one consecutive period.

-- **Notification Object: test**

-- **Notification Window: Retain the default setting.**

-- **Trigger Condition: Retain the default setting.**

Create Alarm Rule

Name:

Description:

Resource Type:

Dimension:

Monitoring Scope:

Select All

InstanceName	ID
☒ ecs-test	8830678-6de3-4084-b076-f102d8c11fa5
☐ as-config-service01-B7YB4YBP	d831570c-4235-45ed-8386-3e4af59e5c87
☐ as-config-service01-AHLOS11I	7148cf88-25ae-4ac5-958a-59ad07408653
☐ as-config-service01-KV38HN9P	4a582cd9-1861-42d3-bb95-9acd23dbae9b

Deselect All

Method:

Alarm Policy

Metric Name	Alarm Policy	Alarm Seve...	Operation
Memory Usage(only for...)	Max	5 minutes	1 consecut...
			>=
		5	%
		Every 5 minutes	

Add Alarm Policy You can add 49 more.

Alarm Notification

Notification Object:

Notification Window: Daily 00:00 - 23:59

Trigger Condition: ☒ Generated alarm ☒ Cleared alarm

You will not be charged for Cloud Eye alarms, but SMN notifications may generate charges based on SMN pricing. [View pricing details](#) for more information.

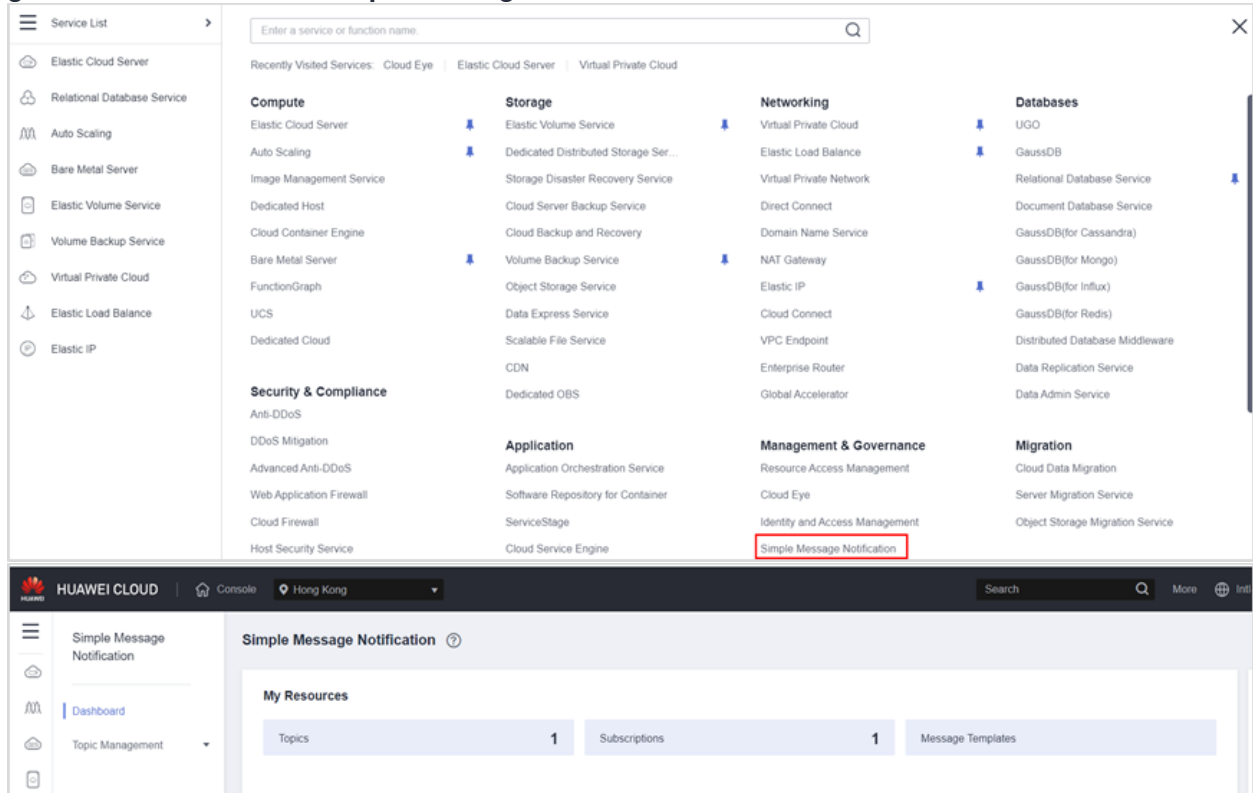
Create

Step 2 Click **Create**. On the **Alarm Rules** page, view the created alarm rule.

Name/ID	Resource Type	Monitored Object	Alarm Policy	Status	Notification Gro...	Alarm Ma...	Alarm Masking D...	Operation
☐ alarm-test2 at1669174766708...	Elastic Cloud Server	ECSs Specific resources	Trigger an alarm if Memory Usage(only for Windows ECSs) Max. >= 5% for 1 consecutive periods of 5 minutes. Trigger an alarm every 5 minutes again if the alarm persists.	Enabled	test	Not mask...		View Details Modify More

3.2 Using SMN

Step 1 Log in to the Huawei Cloud console, click **Service List** in the upper left corner, and choose **Management & Governance > Simple Message Notification**.



Step 2 Choose **Topic Management > Topics**, create Topic

-- **Topic name:test**

Create Topic

★ Topic Name

?

The name cannot be changed after the topic is created.

Display Name

?

Tag

It is recommended that you use TMS's predefined tag function to add the same tag to different cloud resources.
To add a tag, enter a tag key and a tag value below.

10 tags available for addition.

OK

Cancel

locate a topic, and add a subscription to the topic.

-- **Protocol: Email**

-- **Endpoint: Enter an email address.**

Simple Message Notification

Dashboard

Topic Management

Topics

Subscriptions

Message Templates

Topics

Enter a name

Search by Tag

Create Topic

Name	URN	Display Name	Operation
test	urn:snn:ap-southeast-1:0965dd550107a2ff9c00a370c015:test	--	Publish MessageAdd SubscriptionMore

Add Subscription

Topic Name

test

★ Protocol

Email

★ Endpoint ?

Endpoints

com

Description

+ Add Endpoint

Batch Add Endpoints

OK

Cancel

Step 3 Check whether a confirmation email is sent to the entered email address.

If yes, click the **Confirm Subscription** link contained in the email.

Dear Sir or Madam,

Welcome to Simple Message Notification (SMN) service.

You are invited to subscribe to the topic:
urn:smn:ap-southeast-1:09859dfddd00107a2ff9c00a37f0c015:test

After confirmation, you will receive messages posted to this topic via email. Instructions on how to cancel notifications will be included in these messages.

Click the following URL to confirm your subscription: (If you don't want to subscribe to this topic, please ignore this email)

[Confirm Subscription](#)

This URL is valid only within 48 hours.

This is an automatically generated email. Please do not reply.
 Official Website: <https://www.huaweicloud.com/intl/en-us>

Step 4 View the subscription status on the **Subscriptions** page.

Simple Message Notification

Dashboard

Topic Management

Topics

Subscriptions

Message Templates

Subscriptions ?

Request Confirmation

Delete

All protocols

All statuses

Enter an endpoint

Subscription URN	Protocol	Endpoint	Description	Topic Name	Status	Operation
☐ urn:smn:ap-southeast-1:09859dfddd00107a2ff9c00a37f0c015:test.3...	Email		--	test	Confirmed	Request Confirmation Delete

Step 5 Run the following command on the ECS:

```
cd ~
sh test.sh
```

Step 6 Wait for about 5 minutes. Check whether an alarm notification for the CPU usage is sent to the email address you entered.

Dear hwstaff_intl_00256854:

The CPU Usage of Elastic Cloud Server-ECSs "ecs-test" has triggered an alarm, You can log in to the [Cloud Eye console](#) to view details.

Alarm Severity: Major

Alarm Object: Region: CN-Hong Kong

Resource Type: Elastic Cloud Server-ECSs

Resource Name: ecs-test

Resource ID: 8fd39b78-6de3-4084-b076-f102d8cf1fa5

Private IP Address: 192.168.0.25

Public IP Address: 190.92.232.222

Alarm Policy: Trigger an alarm if CPU Usage Max. $\geq 5.00\%$ for 1 consecutive periods of 5 minutes

Current Data: 35.87%

Occurred At: 11 23, 2022 14:40:21 GMT+08:00

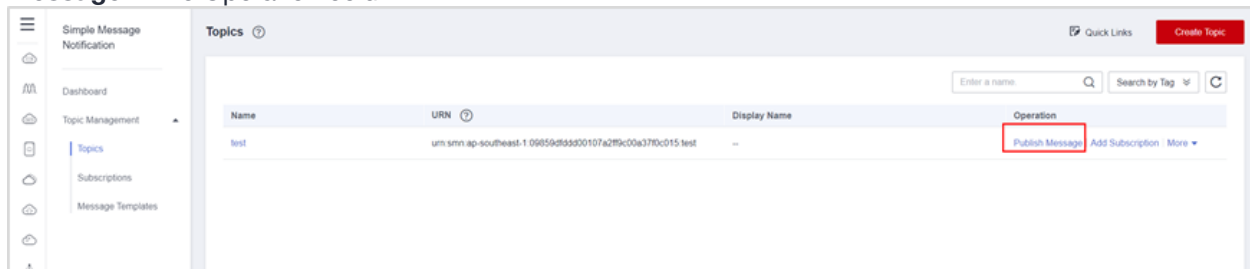
Alarm Data: 11 23, 2022 14:30:00 GMT+08:00 CPU Usage = 35.87%

Alarm Rule Name: alarm-test (ID: al16691730635379Nrk5aWMV)

Alarm Rule Description: --

Monitoring Scope: Specific resources

Step 7 On the SMN console, choose **Topic Management > Topics**, locate the topic, and click **Publish Message** in the Operation column.



Step 8 In the displayed Publish Message dialog box, configure the parameter as follows:

-- **Subject:** cpu

-- **Message Format:** Text

-- **Message:** CPU warning

